PLANT WATERING SYSTEM

CSCI 3515 Final Project

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BACKGROUND

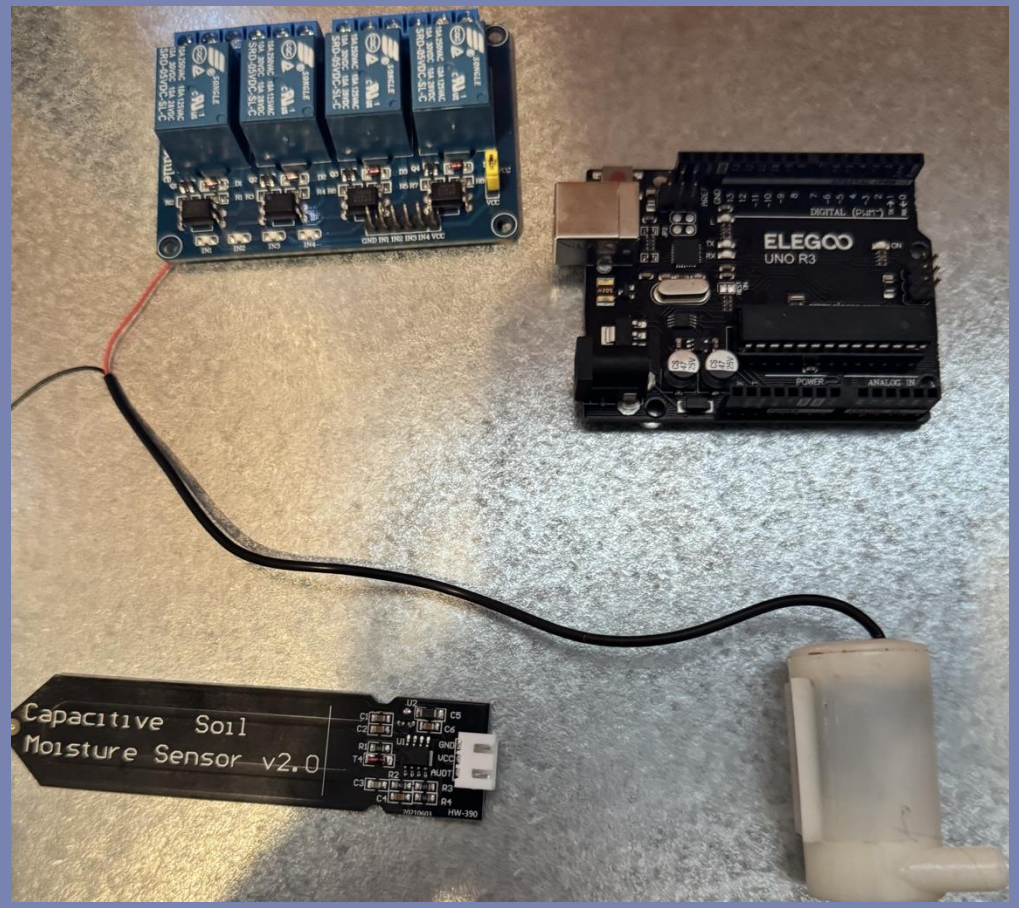
- Three Bonsais: one Small Juniper and two Dwarf Jades.
- Away from tech and school, I invest a lot of time into care for my plants. During busier weeks, sometimes I forget to water.
- I had a Bonsai (Snow Rose) that died in January from underwatering. Some plants are less resilient to watering imbalances.
- I felt bad about this plant death which inspired the idea for this project.

INSPIRATION

- Several of us plant parents have unfortunately let our plants go thirsty at some point, resulting in a few casualties.
- Our goal was to automate the watering process to avoid both overwatering and underwatering, especially for people with busy or unpredictable schedules.
- By making plant care more hands off, this system helps save money by reducing the number of dead plants and frees up time for users who already have a lot on their plate.

COMPONENTS

- ❑ Arduino Uno R3 Microcontroller
- ❑ DHT11 Temperature and Humidity Sensor
- ❑ Capacitive Soil Moisture Sensor v2.0
- ❑ Breadboard and jumper wires
- ❑ LEDs as status indicators
- ❑ Fan for environmental humidity control
- ❑ Tubing for water delivery
- ❑ 4-Channel Relay Module
- ❑ DC Water pump
- ❑ Lorenz ‘s Bonsai Tree

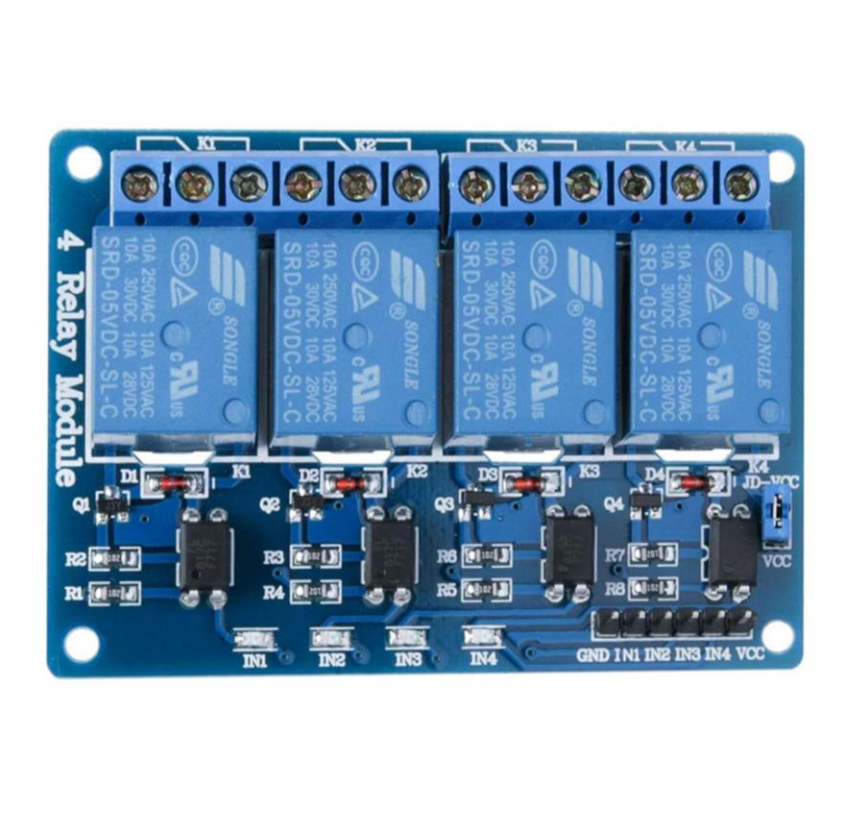


HOW IT WORKS

- Soil moisture sensor checks soil
- If soil is too dry, the Arduino communicates with DC pump to water plant
- LEDs show moisture status:
  - Green – soil moisture level is good
  - Yellow – soil needs watering soon
  - Red – soil needs watering asap
  - Blue – DC pump is ON + actively watering
- The DC pump will turn ON/OFF based on data collected from moisture sensor

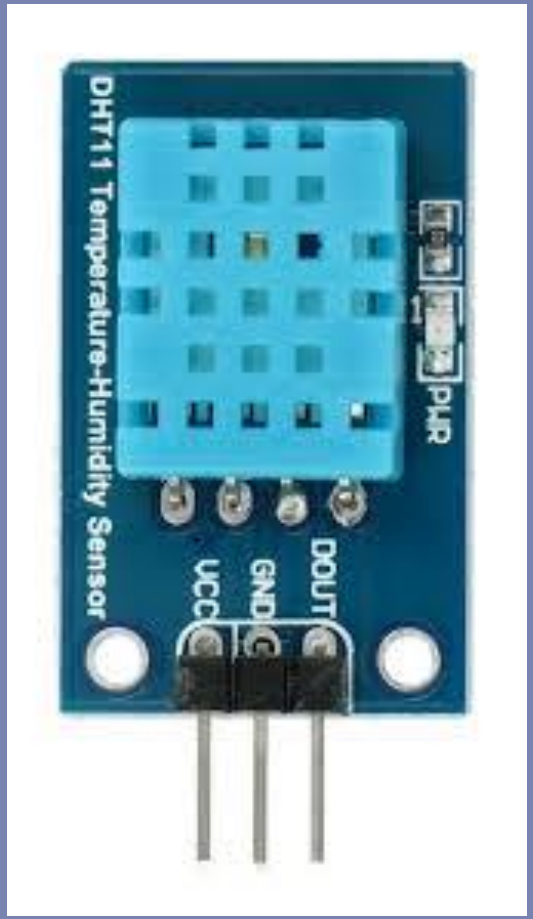
4-CHANNEL RELAY MODULE

- Allows use of low-powered device like Arduino to power high-powered devices like our pump
- Relay acts as a switch that uses signals to open or close a separate circuit to handle power
- This creates electrical isolation to prevent damage to the board



TEMPERATURE & HUMIDITY SENSOR

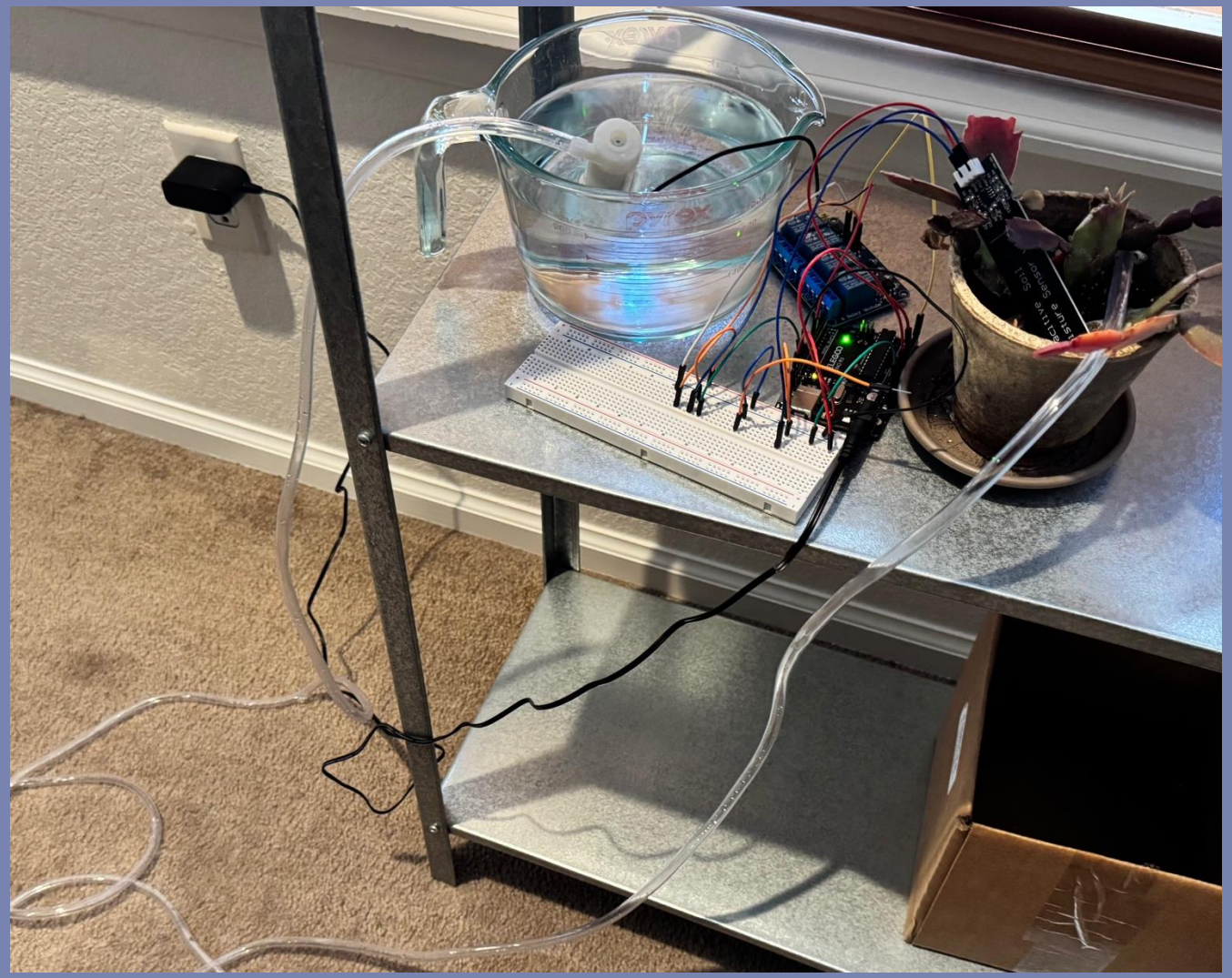
- Measures air temperature + relative humidity
- Sensor communicates with Arduino via digital signal
- Arduino requests/receives temperature and humidity values
- Our system calculates a heat index and turns on fan if it is too hot/humid for the plant

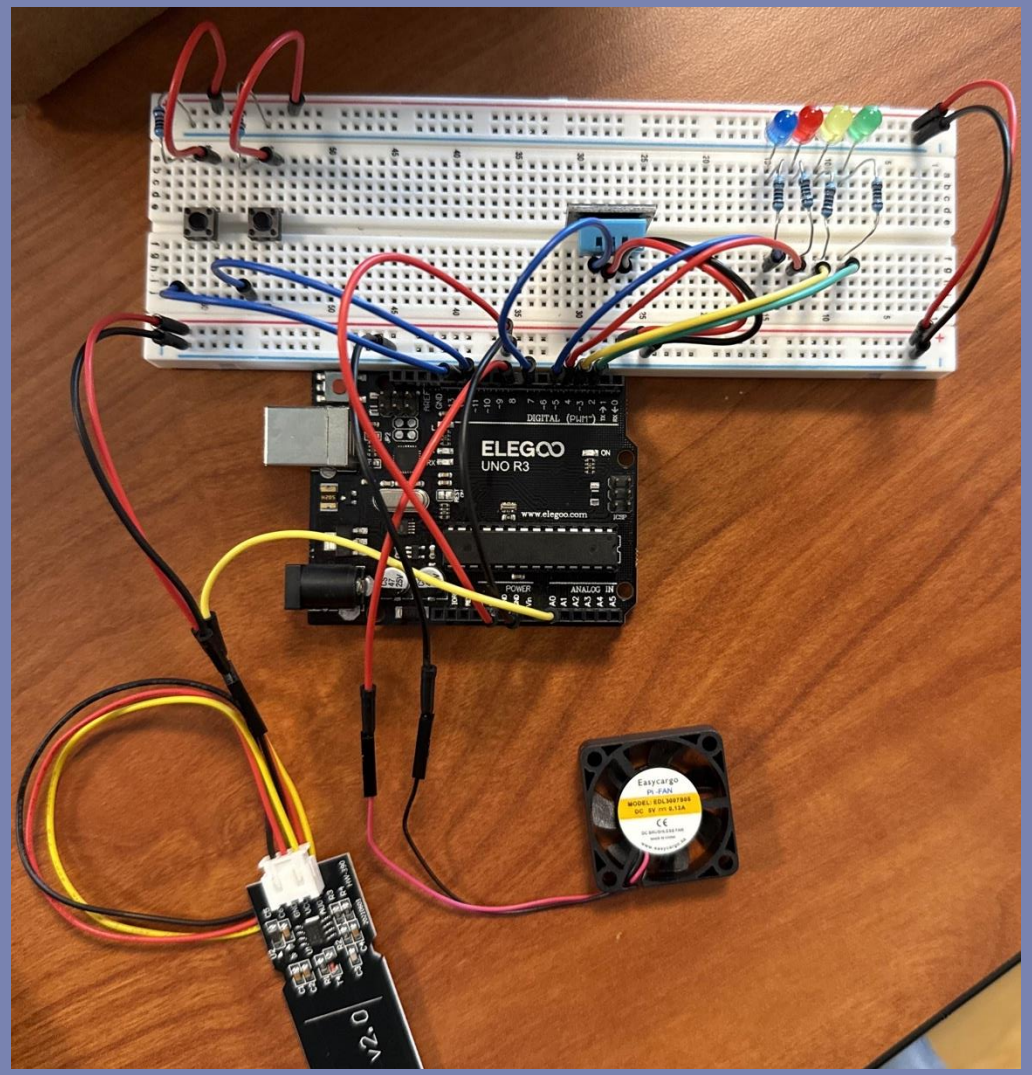


CAPACITIVE SENSOR

- Sensor detects changes in capacitance (ability to store electric charge) caused by soil moisture
- Sends analog signal to Arduino
  - Dry soil: lower voltage
  - Wet soil: higher voltage
- Arduino compares value to dryness threshold to decide whether to turn on water pump
- Capacitance sensors have no exposed metal which lowers likelihood of corrosion and increases longevity & durability.

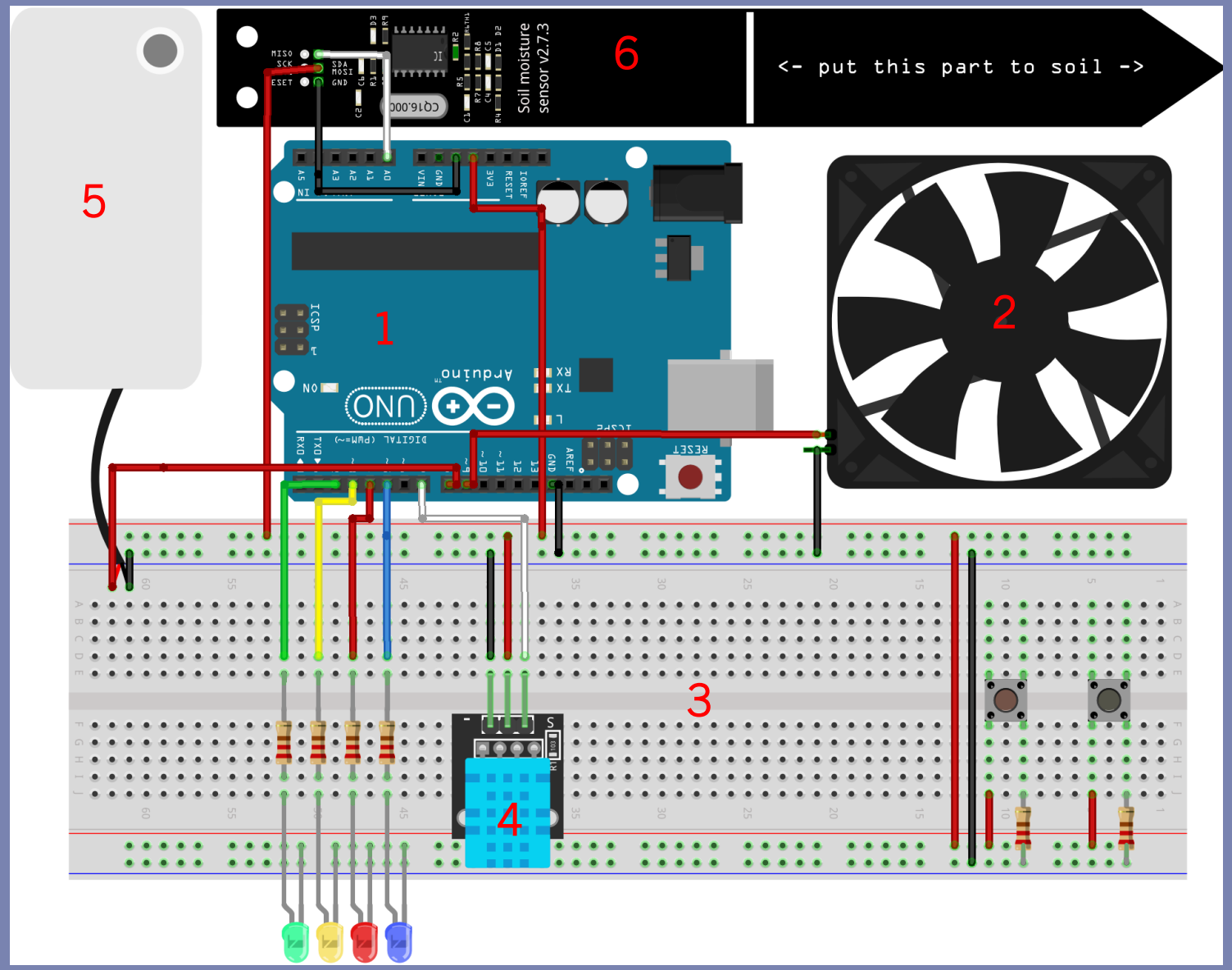
Project Photos





Fritzing Diagram with Key

1: Arduino Uno
2: Fan
3: 830 Tie-points Breadboard
4: DH11 Temperature and Humidity Sensor
5: DC Water Pump
6: Capacitive Soil Moisture Sensor v2.0



ANY QUESTIONS?

DemonstrationVideo.mov